MIL-S-17743E
16 February 1973
SUPERSEDING
MIL-S-17743D
23 July 1969

MILITARY SPECIFICATION

STRETCHERS, STEEL STRAPPING, HAND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand operated stretchers for tensioning steel strapping on bundles, boxes, cartons, crates, pallets, and car loading.

* 1.2 Classification. Stretchers shall be of the following types, styles, classes, and sizes as specified (see 6.2).

Type I - Feed wheel, unlimited takeup (for open or thread-on seals).

Size - 3/8 - 1/2
Size - 5/8 - 3/4
Size - 3/4 - 1-1/4

Type II - Rack and gear takeup.

Style 1 - For open or thread-on seals

Size - 3/8 - 3/4 (unlimited takeup)
Size - 3/8 - 3/4 (limited takeup)
Size - 3/4 - 1-1/4 (limited takeup)

Style 2 - For push-type seals

Size - 3/4 - 1-1/4 (unlimited takeup)

FSC 3540

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Type III - Drum limited takeup (for open or thread-on seals).

Class A - Without shear device

Size - 3/4 - 1-1/4
Size - 1-1/4 - 2

Class B - With built-in shear device

Size - 3/4 - 1/4

(Sizes indicate range of strapping widths accommodated by the stretcher.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel, Strapping, Flat
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Set-up, Paperboard
PPP-F-320	-	Fiberboard; Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier-Materials, Grease-Proof, Waterproofed, Flexible
MIL-B-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

Standard ASTM - E18, Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2 and 6.3).

* 3.2 Materials and components. Materials and components shall be as specified herein and on applicable drawings. Materials not definitely specified shall be of the quality normally used by the manufacturer in his commercial product provided that the completed item complies with all provisions of this specification.

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* 3.3 Design and construction. Type I, II and III stretchers shall be manually operated and shall be capable of taking up and tensioning flat steel strapping of the sizes specified herein and conforming to QQ-S-781, by the reciprocating action of a handle with a ratchet. The stretchers shall be capable of retaining the tension of the strapping as the handle is released. The stretchers shall have means for releasing strapping from the stretchers after the sealing operation has been accomplished. The stretchers shall accomplish limited or unlimited takeup of the strapping as specified (see 1.2).

* 3.3.1 Hardness. Components of the stretchers, when tested in 4.4.1, shall have the following hardness with no evidence of surface cracks.

(1) Feed wheels, push-type noses, gripping dogs, gripping pads and shear blades shall have a through hardness of Rockwell "C" 55 to 60.

(2) Winding drums for strapping, ratchet wheels, pawls and gears, racks, gear carriage, bushings and pins shall have a through hardness of Rockwell "C" 45 to 64 or a case hardness of Rockwell "30-N" 69 to 77 for plain carbon steels and a through hardness of Rockwell "C" 20-25 for alloy steels.

* 3.3.2 Sizes. Type I, II and III stretchers of sizes specified (see 1.2) shall takeup and tension strapping of sizes as indicated in table I.

TABLE I. Strapping sizes for stretchers

Stretcher	Size	Strapping sizes (inches) Width	Thickness
Type I	3/8 - 1/2	3/8 to 1/2	0.010 to 0.023
	5/8 - 3/4	5/8 to 3/4	0.010 to 0.023
	3/4 - 1-1/4	3/4 to 1-1/4	0.025 to 0.035
Type II			
Style 1	3/8 - 3/4	3/8 to 3/4	0.010 to 0.023
	3/4 - 1-1/4	3/4 to 1-1/4	0.028 to 0.050
Style 2	3/4 - 1-1/4	3/4 to 1-1/4	0.028 to 0.050
Type III			
Class A	3/4 - 1-1/4	3/4 to 1-1/4	0.028 to 0.050
	1-1/4 - 2	1-1/4 to 2	0.050
Class B	3/4 - 1-1/4	3/4 to 1-1/4	0.028 to 0.050

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3.3.3 Type I. Type I stretcher (see 6.1.1) shall take up and tension steel strapping in the direction of the coil, by means of a feedwheel under spring tension. The feedwheel, actuated by the ratcheting handle, shall accomplish unlimited takeup by biting into and sliding the strapping on the coil side over the free end of the strapping. The free end shall be held fast by a gripping pad in the base of the stretcher, in combination with pressure exerted by the feedwheel.

3.3.4 Type II. Type II stretchers shall takeup and tension steel strapping in the direction of the coil by means of a gear and rack. Type II, style 1 and 2 stretchers shall accomplish limited and unlimited takeup of the strapping as specified herein.

* 3.3.4.1 Style I. Style 1 stretcher (see 6.1.2) shall be a rack and gear stretcher with unlimited or limited takeup and tensioning of steel strapping.

* 3.3.4.1.1 Unlimited takeup. Style 1 stretcher with unlimited takeup shall have three (3) gripping dogs as follows:

(1) One dog for gripping the free end of the strapping shall be located on the nose end of the stretcher and shall remain fixed.

(2) The remaining two dogs in tandem; one dog located on the nose end of the stretcher shall remain fixed and the second dog located on and moveable with the gear carriage shall grip the coil side of the strapping for accomplishing unlimited takeup and tensioning of the strapping.

* 3.3.4.1.2 Limited takeup. Style 1 stretcher with limited takeup shall have two (2) gripping dogs. One dog located on the nose end of the stretcher shall grip the free end of the strapping and remain fixed. The second dog located on and moveable with the gear carriage shall grip the coil end of the strapping for accomplishing limited takeup and tensioning of the strapping that shall be limited by the length of the rack.

* 3.3.4.2 Style 2. Style 2 stretcher (see 6.1.3) shall have two (2) gripping dogs and a push-type nose. The nose shall bear against a push-type seal threaded with two (2) thicknesses of strapping (the coil side and free end) and held fast by the free end of the strapping bent around and under the seal. The two (2) dogs (one located on the nose end and the other located on the gear carriage of the stretcher) shall both grip the strapping on the coil side and shall operate to provided unlimited takeup and tensioning of the strapping. The stretcher, by means of the push-type nose, shall allow the operator to bend and break off the strapping flush at the seal after the sealing operation has been accomplished.

3.3.5 Type III, class A. Type III, class A stretchers (see 6.1.4) shall have a body, a slotted drum for mounting of one free end of a cut-to-length strapping, a ratchet with handle for rotating the drum and a

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gripping dog located on the body for gripping and holding fast the remaining free end of the strapping. Takeup and tensioning of the strapping shall be limited by the number of laps of strapping that the drum will accommodate.

3.3.6 Type III, class B. Type III, class B stretcher shall be the same as class A except that class B shall have a shear device for cutting off excess strapping after the sealing operation has been accomplished.

* 3.4 Performance. Type I, II, and III stretchers shall operate when tested in 4.4.2 as follows:

(1) The stretcher shall be threaded and unthreaded with no interference or obstructions from threading components.

(2) Each type, style or class and size of stretcher shall; (1) takeup, tension and retain the tension of the applicable maximum size strapping in accordance with applicable characteristics in 3.3.2 thru 3.3.6; and (2) apply the minimum load applicable to the maximum size strapping as indicated in table II, with no binding of stretcher components and no visual slippage of strapping and deformation, fracture, or breakage of components.

TABLE II. Loads (minimum)

Strapping Width (inches)	Load (pounds)
1/2 to 3/4 (light duty strapping)	75 percent minimum breaking strength
3/4 (heavy duty strapping)	75 percent minimum breaking strength
1-1/4 to 2 (heavy duty strapping)	50 percent minimum breaking strength

3.5 Repair and maintenance. Stretchers shall have means for adjustment, service, and replacement of components.

3.6 Marking for identification. The stretchers shall be marked for identification in accordance with manufacturer's standard practice and shall include the strapping size range for stretcher use. The marking shall be permanent and legible on a corrosion-resistant metal plate or metallic decalcomania.

3.7 Finish. Finish of the stretcher shall be in accordance with the manufacturer's commercial practice. The stretcher shall be clean and free from burrs, slivers and rust areas.

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* 3.8 Workmanship. The stretchers shall have no binding or misalignment of components. Components shall not be loose, bowed, dented or fractured.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article. When a first article is required, it shall be examined in accordance with table III, dimensions specified and tested in accordance with 4.4.1 and 4.4.2. The presence of any visual defects, dimensions not specified, or failure to pass any test shall be cause for rejection of the first article.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Inspection of components and materials. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End product inspection. A lot shall consist of all the stretchers of one type, style or class, and size offered for acceptance at one time. The sample unit for this inspection shall be one complete stretcher.

4.3.2.1 Visual examination. The examination of the end item shall be made for the defects specified in table III. The inspection level shall be level II with an Acceptable Quality Level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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* TABLE III. Examination of visual defects

Examine	Defect	Classification Major	Minor
Finish	Not in accordance with manufacturer's commercial practice	X	
	Unit not clean or free from burrs, slivers or rust areas		X
Construction and workmanship	Unit with binding or misalignment of component	X	
	Components loose		X
	Bowed, dented, or fractured	X	
Identification marking	Missing, incomplete, or not legible		X

4.3.2.2 Dimension examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any nonconformance shall be considered a defect. The inspection level shall be S-2 and an AQL of 6.5 expressed in terms of defects per hundred units.

* 4.3.2.3 End item testing. When a first article is not required, the initial production unit shall be tested as specified in 4.4.1 and 4.4.2. Failure to pass any test shall be cause for rejection of the initial unit and of the items produced. Each stretcher shall be tested as specified in 4.4.2. Any nonconformance shall be cause for rejection of the item.

4.3.3 Examination of preparation for delivery. An examination will be made to determine that preservation, packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table IV. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Markings	Omitted, incorrect, illegible, of improper size location, sequence, or method of application
Preservation	Preservative improperly applied or missing

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TABLE IV. Examination of preparation for delivery (cont'd)

Examine	Defect
Materials	Component missing, damaged, or otherwise defective
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, bulging or distortion of containers
Weight	Weight of contents exceeds requirements

4.4 Tests.

4.4.1 Hardness test. Each heat treated and case hardened component shall be tested as follows:

(1) The components shall be checked for surface cracks using a glass of minimum 5X magnification. Inability of components to meet surface crack requirements of 3.3.1 shall constitute failure to pass this test.

(2) The average hardness of the components shall be obtained from a minimum of 3 determinations in accordance with ASTM E18.

Inability of components to meet hardness requirements as specified in 3.3.1 shall constitute failure to pass this test.

* 4.4.2 Performance test. Stretcher performance shall be tested as follows:

(1) The stretcher shall be threaded with the applicable maximum size strapping simulating the application of strapping on a package and checked for ease of threading. Inability of the stretcher to meet threading requirements of 3.4 (1) shall constitute failure of this test.

(2) The stretcher except type II, style 2, threaded with the applicable maximum size strapping, shall be attached to a tension measuring apparatus and shall be operated to takeup and tension the strapping until the specified load in table II has been reached. Type II, style 2 stretcher shall be tested in a similar manner with the applicable maximum size strapping except that the nose end instead of butting against a closed seal, shall butt against a fixed object. Extension of the stretcher handles to meet the requirements is acceptable. Inability of the stretchers to meet the takeup, tensioning and load requirements as specified in 3.4 (2) shall constitute failure to pass this test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. All exposed, uncoated ferrous metal surfaces of each stretcher shall be cleaned process C-1 of MIL-P-116 and thoroughly dried. Surfaces, cleaned and dried as specified, shall be coated with type P-2 or P-6 preservative of MIL-P-116 and wrapped in greaseproof barrier material conforming to type II, grade A, class 2 of MIL-B-121. Each stretcher shall then be packaged in a snug fitting paperboard box conforming to PPP-B-566 or PPP-B-676 or in a fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The stretcher shall be immobilized within the box by means of corrugated fiberboard inserts or cushioning material. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C. Stretchers shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during known favorable conditions of shipment, handling, and storage from the supply source to the receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Stretchers, preserved and packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification. The weight of contents shall not exceed 65 pounds. Alternatively containers, constructed, closed, and strapped in accordance with class 2 style 2 or 4 of PPP-B-621 may be used. When containers conforming to PPP-B-621 are used they shall be provided with a sealed case liner conforming to type I or II grade B or C style optional of MIL-L-10547.

5.2.2 Level B. Unless otherwise specified (see 5.2.2.1), stretchers preserved and packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, type CF, variety SW or type SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with method II of the appendix of the container specification. The weight of contents shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2) shipping containers shall be made from grade V3c or V3s fiberboard. The box shall be fabricated in accordance with PPP-B-636 and closed in accordance with the weather resistance requirements of the appendix of PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320.

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5.2.3 Level C. Stretchers, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The stretchers are used for applying flat steel strapping on bales, boxes, bundles, cartons, crates, and pallets. The heavier sizes are also used for securing material and equipment in freight cars, ships, and trucks for shipment.

6.1.1 Type I stretcher is suitable for general use on lubricated strapping, uncoated or coated with an organic finish, direct from the coil.

6.1.2 Type II, style 1 stretcher is suitable for use on strapping with any class of finish, direct from the coil, on narrow or irregular packages where high tension in the strapping is desired.

6.1.3 Type II, style 2 stretcher is suitable for use on strapping with any class of finish, direct from the roll, on narrow, round or irregular packages and bundles with little or no compression.

6.1.4 Type III, class A and B is suitable for use on cut-to-length heavy duty strapping with any class of finish.

* 6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements

(a) Title, number and date of this specification.
(b) Type, style or class and size (see 1.2).
(c) First article. When a first article is required (see 3.1, 4.2 and 6.3).
(d) Applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).
(e) Whether weather-resistant fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

* 6.2.2 Contract data requirements. Any requirement for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

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* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) were made from previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Stretchers, Steel Strapping, Hand MIL-S-17743E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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